CHEMICAL REACTIONS!

Hot vs Cold Magic • Grade 4 Fun Science

Rub your hands together fast! What do you feel? They get super warm! Now think about placing an ice pack on your knee—it feels freezing cold! Today, we are going to become **Science Detectives** to unlock the secret of why some chemical reactions make things HOT and others make them COLD!

1. What is a Chemical Reaction?

A **chemical reaction** happens when starting materials (called *Reactants*) change completely into something totally new (called *Products*)!

Reactants ➔ Products (A New Superhero Substance is Born!)

- **Cooking Roti on a Tawa:** Soft dough turns into a delicious cooked roti! (New substance!)
- **Rusting Gate:** Shiny iron turns into reddish rust after rain.
- **NOT a reaction:** Ice melting into water (It's still just water!), or dissolving sugar in milk.

2. The Two Secret Superpowers: EXO & ENDO!

Every chemical reaction plays with heat energy. Use these super easy memory tricks to spot them instantly:

EXOTHERMIC

EXO = EXIT!

Heat **EXITS** the reaction and shoots out into the room! The surroundings get **HOT!**

Clue Words: gets hot, flame, burning, gives off heat.

Fun Examples:

- Burning Diwali sparklers & crackers
- Coal burning in a winter sigri
- Cooking dal on a gas stove

ENDOTHERMIC

ENDO = ENTER!

Heat **ENTERS** and gets sucked inside! The surroundings get **COLD!**

Clue Words: gets cold, absorbs heat, cools down.

Fun Examples:

- Water staying cool in an earthen matka
- Mixing baking soda + vinegar (feels ice-cold!)
- Plants taking in sunlight for food

3. Student Quick Summary Cheat-Sheet

Reaction Type	Memory Trick	Heat Flow Direction	How It Feels	Everyday Example

Exothermic	EXO = EXIT	Heat shoots OUT	**HOT!** 🔥	Burning a campfire or candle
Endothermic	ENDO = ENTER	Heat gets sucked IN	**COLD!** ❄️	Ice pack on a bumped knee

STUDENT SCIENCE CHALLENGE!

Practice Questions • Test Your Superpowers

Instructions for Young Scientists: Answer all the questions below carefully. Use your clue words (gets hot, feels cold, burns, absorbs heat) to solve each puzzle!

QUESTION 1

What happens during a chemical reaction?

A) Things stay completely identical

B) Starting materials change into a brand new substance

C) Water turns into ice and back

D) Sugar disappears forever

QUESTION 2

Fill in the blanks using your memory tricks (EXO = EXIT, ENDO = ENTER):

a) An ______________ reaction releases heat, so the glass gets **HOT**!

b) An ______________ reaction absorbs heat, so the container gets **COLD**!

QUESTION 3

Classify each event below! Write "EXOTHERMIC" or "ENDOTHERMIC" next to each:

a) Lighting a Diwali diya or candle: ________________________

b) Mixing baking soda and vinegar in a cup (feels cold): ________________________

c) Placing an ice pack on an injured ankle: ________________________

d) Wood burning in a Lohri bonfire: ________________________

QUESTION 4

Why does water stay nice and cool inside a traditional clay matka (pot) during hot Indian summers? Explain using what you learned about heat flow!

QUESTION 5 (BONUS SPY CHALLENGE)

Look around your kitchen or home! Write down ONE real-life chemical reaction you have seen that gives off heat or feels cold. Is it Exothermic or Endothermic?